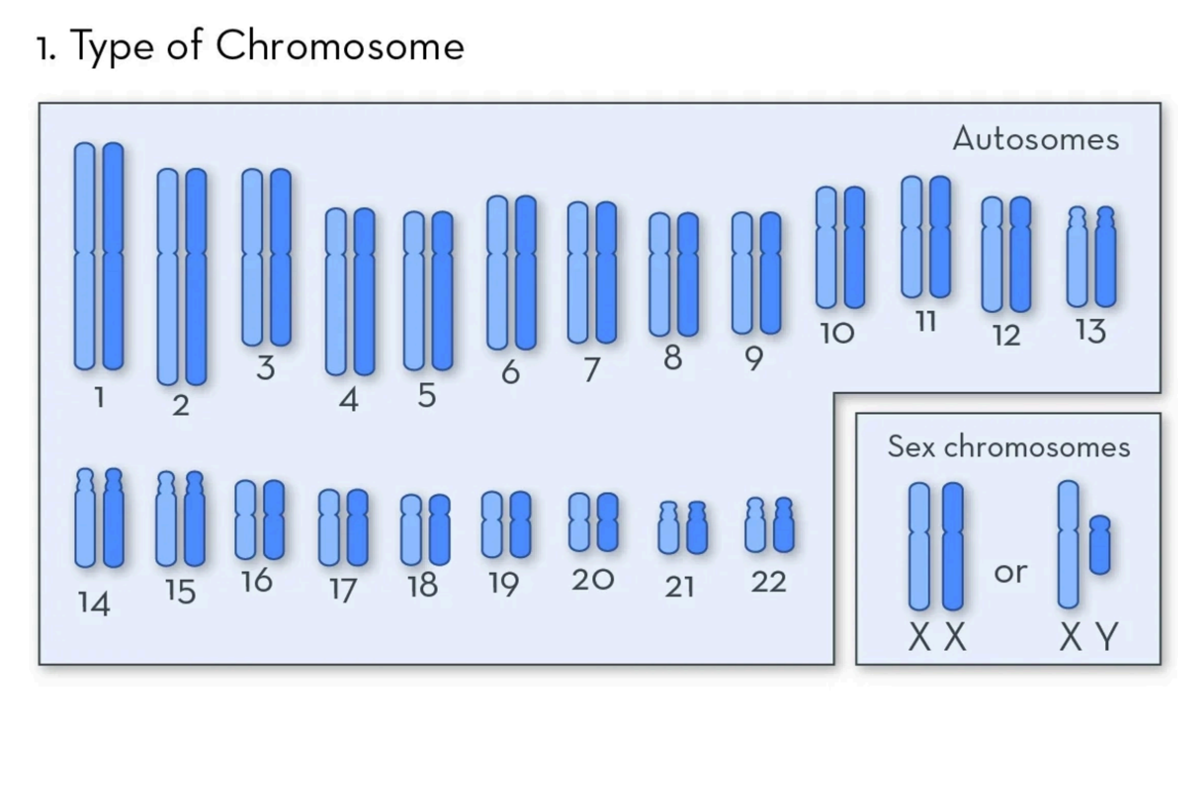
5.1 Modes of Inheritance

- What are autosomes?
  - The 22 non sex chromosomes



- When is a genetic disorder described as autosomal dominant?
  - When you only need one copy of the allele to cause the relevant change in phenotype.
- What gene is responsible for controlling brown eye colour?
  - OCA-2
- Recall the three types of autosomal dominant disorders.
  - Gain-of-function - mutation causes production of newly-functioning protein
  - Dominant negative - mutation causes protein product to interact differently with other molecules
  - Haploinsufficient - one copy of a gene is inactivated or deleted and the remaining functional copy of the gene is not adequate to produce the needed gene product to preserve normal function.

- What is meant by the term "haploinsufficiency?"
    - There is a loss of one allele, and the remaining allele cannot produce enough protein to ensure regular function (i.e collagen)
- Describe the mutation that occurs in Huntington's disease.
    - A repeat expansion of the CAG (glutamine) residue
- How does this mutation affect sufferers?
    - The abnormal intracellular Huntington protein gains a pathological function and can lead to cell death
- What happens in hypertrophic cardiomyopathy and how does it affect sufferers?
    - It is abnormal thickening of the heart muscles
    - It can lead to a reduced volume of the heart chambers and cause sudden cardiac death
- What type of mutation typically causes recessive disorders?
    - Loss-of-function
- What is a loss-of-function mutation?
    - A type of mutation in which the **altered gene product lacks the molecular function** of the wild-type gene
- What is consanguinity?
    - The fact of being descended from the same ancestor.
- What effect does consanguinity have on probability of inheriting disorders?
    - It increases the risk of inheriting **autosomal recessive disorders only.**
- Describe the mutation that causes cystic fibrosis.
    - Mutations occur on the gene that encodes for the chloride ion channel (chromosome 7)
- What does this mutation lead to, and what type of mutation is it?
    - Defective chloride ion channel
    - This is a **loss-of-function mutation**
- Recall a structural distinction between X and Y chromosomes.
    - X chromosomes are much larger than Y chromosomes
    - This means they contain around 1000-1300 genes but Y only contain about 150 genes

▼ Why are X-linked disorders effectively dominant in men?

- Men only have one copy of the X-chromosome so they only need one copy of an allele for a phenotype to be expressed through the sex chromosomes.

▼ Why are sons of fathers with X-linked dominant disorders not affected?

- Sons only inherit the Y-chromosome from their fathers
- They inherit their X-chromosome from their mothers.

▼ Where do we inherit our mitochondria from?

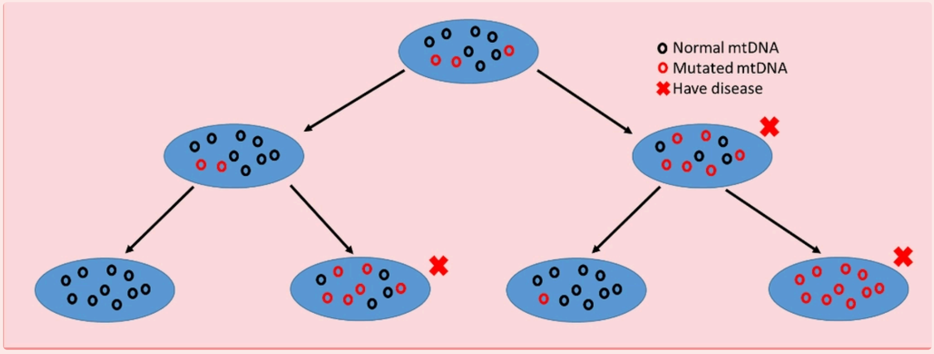
- From our mothers.

▼ What characteristic of mitochondrial conditions makes them difficult to predict within a family?

- **Heteroplasmy**
- This leads to a high level of uncertainty as to which symptoms will be exhibited, how early the onset will be etc.

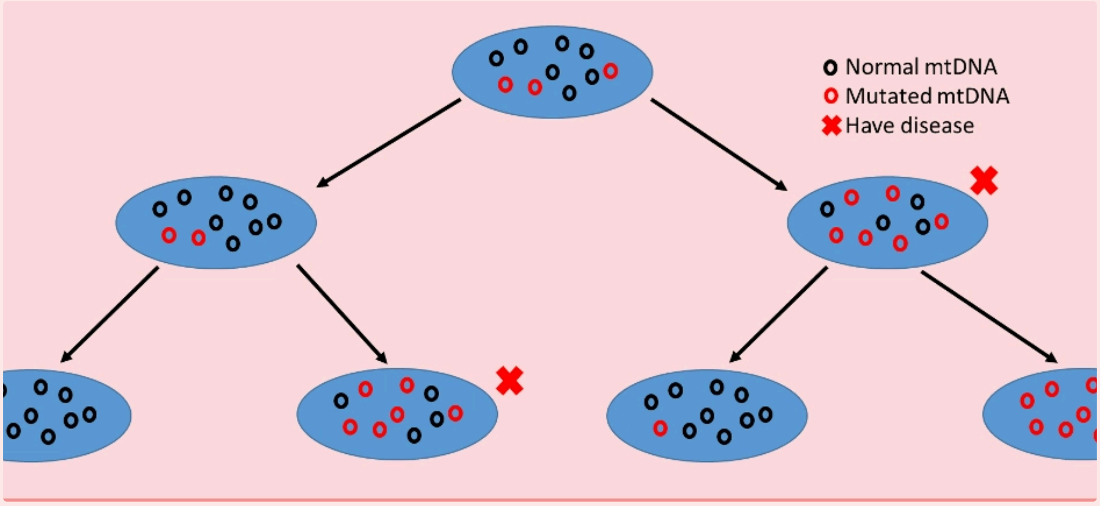
▼ What is heteroplasmy?

- It is when there is **more than one type of mitochondrial DNA (mtDNA) within one cell**



▼ How does mitochondrial replication increase mitochondrial disease variability?

- Mitochondria undergo what is known as binary fission and this leads to the gain or loss of mutated genes across generations in replication
- There are also **multiple mitochondria** within cells with different genes, so it is **difficult to predict the number of mutated mitochondria** that will be present in an offspring
    - This is because mitochondrial replication occurs as **random segregation**



▼ NOTE FOR READING PEDIGREE DIAGRAMS.